

JEE-MAINS-2020-JAN-8-SHIFT-1

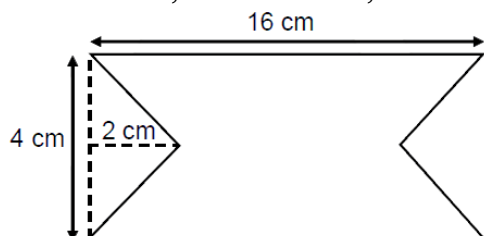
PHYSICS

Question 1: Consider a uniform rod of mass $M=4m$ and length l pivoted about its centre. A mass m moving with velocity v making angle $\theta = \frac{\pi}{4}$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod - mass system just after the collision is:

Options:

- (a) $\frac{4}{7} \frac{v}{l}$
- (b) $\frac{3\sqrt{2}}{7} \frac{v}{l}$
- (c) $\frac{3}{7} \frac{v}{l}$
- (d) $\frac{3}{7\sqrt{2}} \frac{v}{l}$

Question 2: At time $t = 0$ magnetic field of 1000 Gauss is passing perpendicularly through the area defined by the closed loop shown in the figure. If the magnetic field reduces linearly to 500 Gauss, in the next 5 s, then induced EMF in the loop is



Options:

- (a) $28 \mu\text{V}$
- (b) $36 \mu\text{V}$
- (c) $48 \mu\text{V}$
- (d) $56 \mu\text{V}$

Question 3: Proton with kinetic energy of 1 MeV moves from south to north. It gets an acceleration of 10^{12} m/s^2 by an applied magnetic field (west to east). The value of magnetic field: (Rest mass of proton is $1.6 \times 10^{-27} \text{ kg}$)

Options:

- (a) 7.1 mT
- (b) 71 mT
- (c) 0.71 mT
- (d) 0.071 mT

Question 4: In finding the electric field using Gauss law the formula $|\vec{E}| = \frac{Q_{in}}{\epsilon_0 |A|}$ is

applicable. In the formula ϵ_0 is permittivity of free space, A is the area of Gaussian surface and q_{enc} is charge enclosed by the Gaussian surface. This equation can be used in which of the following situation?

Options:

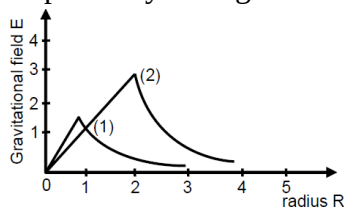
- (a) Only when the Gaussian surface is an equipotential surface.
- (b) Only when E = constant on the surface.
- (c) Only when the Gaussian surface is an equipotential surface and E is constant of the surface
- (d) For any choice of Gaussian surface

Question 5: Effective capacitance of parallel combination of two capacitors C_1 and C_2 is $10 \mu\text{F}$. When these capacitors are individually connected to a voltage source of 1 V, the energy stored in the capacitor C_2 is 4 times that of C_1 . If these capacitors are connected in series, their effective capacitance will be

Options:

- (a) $4.2 \mu\text{F}$
- (b) $8.4 \mu\text{F}$
- (c) $3.2 \mu\text{F}$
- (d) $1.6 \mu\text{F}$

Question 6: Consider two solid spheres of radii $R_1 = 1\text{m}$ $R_2 = 2\text{m}$ and masses M_1 and M_2 , respectively. The gravitational field due to sphere 1 and 2 are shown. The value of M_1/M_2 is



Options:

- (a) $\frac{2}{3}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{2}$
- (d) $\frac{1}{6}$

Question 7: A leak proof cylinder of length 1 m, made of a metal which has very low coefficient of expansion is floating vertically in water at 0°C such that its height above the water surface is 20 cm. When the temperature of water is increased to 4°C , the height of the

cylinder above the water surface becomes 21 cm. The density of water at $T = 4^\circ\text{C}$, relative to the density at $T = 0^\circ\text{C}$ is close to

Options:

- (a) 1.03
- (b) 1.04
- (c) 1.26
- (d) 1.01

Question 8: The magnifying power of a telescope with tube length 60 cm is 5. What is the focal length of its eye piece?

Options:

- (a) 40 cm
- (b) 20 cm
- (c) 10 cm
- (d) 30 cm

Question 9: A particle of mass m is fixed to one end of a light spring having force constant k and unstretched length l . The other ends is fixed. The system is given an angular speed ω about the fixed end of the spring such that it rotates in a circle in gravity free space. Then the stretch in the spring is

Options:

- (a) $\frac{ml\omega^2}{k + m\omega}$
- (b) $\frac{ml\omega^2}{k + m\omega^2}$
- (c) $\frac{ml\omega^2}{k - m\omega^2}$
- (d) $\frac{ml\omega^2}{k - \omega m}$

Question 10: The graph which depicts and results of Rutherford gold foil experiment with α -particles is

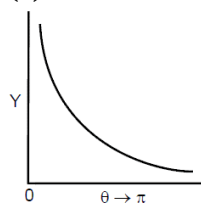
θ - Scattering angle

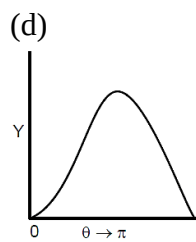
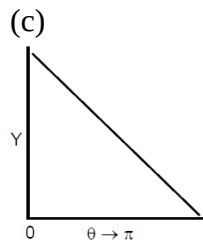
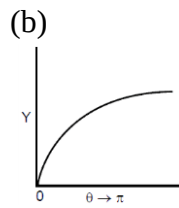
Y: Number of scattered α -particles detected

(Plots are schematic and not to scale)

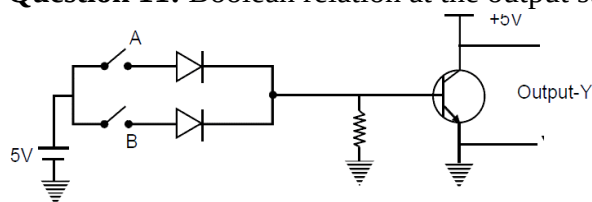
Options:

(a)





Question 11: Boolean relation at the output stage-Y for the following circuit is



Options:

- (a) $\overline{A} + \overline{B}$
- (b) $\overline{A} \cdot \overline{B}$
- (c) $A \cdot B$
- (d) $A + B$

Question 12: When photon of energy 4.0 eV strikes the surface of a metal A, the ejected photoelectrons have maximum kinetic energy T_A eV and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photon of energy 4.50 eV is $T_B = (T_A - 1.5)$ eV. If the de-Broglie wavelength of these photoelectrons $\lambda_B = 2\lambda_A$, then the work function of metal B is

Options:

- (a) 2 eV
- (b) 3 eV
- (c) 1.5 eV
- (d) 4 eV

Question 13: The critical angle of a medium for a specific wavelength, if the medium has relative permittivity 3 and relative permeability $4/3$ for this wavelength, will be

Options:

- (a) 15°
- (b) 30°
- (c) 60°
- (d) 45°

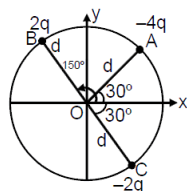
Question 14: Consider a solid sphere of radius R and mass density

$\rho(r) = \rho_0 \left(1 - \frac{r^2}{R^2} \right), 0 < r \leq R$. The minimum density of a liquid in which it will float is

Options:

- (a) $\frac{\rho_0}{3}$
- (b) $\frac{2\rho_0}{3}$
- (c) $\frac{\rho_0}{5}$
- (d) $\frac{2\rho_0}{5}$

Question 15: Three charged particles A, B and C with charges $-4q$, $2q$ and $-2q$ are present on the circumference of a circle of radius d . The charged particles A, C and centre O of the circle formed an equilateral triangle as shown in figure. Electric field at O along x-direction is:



Options:

- (a) $\frac{\sqrt{3}q}{\pi\epsilon_0 d^2}$
- (b) $\frac{\sqrt{3}q}{4\pi\epsilon_0 d^2}$
- (c) $\frac{3\sqrt{3}q}{4\pi\epsilon_0 d^2}$
- (d) $\frac{2\sqrt{3}q}{\pi\epsilon_0 d^2}$

Question 16: The dimension of stopping potential V_0 in photoelectric effect in units of Planck's constant 'h', speed of light 'c' and Gravitational constant 'G' and ampere A is

Options:

- (a) $h^{-2/3} c^{-1/3} G^{4/3} A^{-1}$

- (b) $h^{2/3} G^{5/3} c^{1/3} A^{-1}$
 (c) $h^{1/3} G^{2/3} c^{1/3} A^{-1}$
 (d) $h^{2/3} G^{3/2} c^{1/3} A^{-1}$

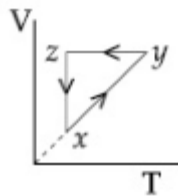
Question 17: The length of a potentiometer wire is 1200 cm and it carries a current of 60 mA. For a cell of emf 5 V and internal resistance of 20Ω , the null point on it is found to be at 1000 cm. The resistance of whole wire is

Options:

- (a) 60Ω
 (b) 120Ω
 (c) 100Ω
 (d) 80Ω

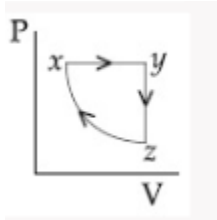
Question 18: The thermodynamic cycle $xyzx$ is shown on a V-T diagram.

The P-V diagram that best describes this cycle is : (Diagrams are schematic and not to scale)

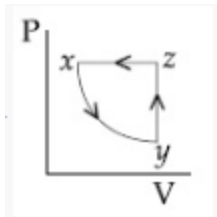


Options:

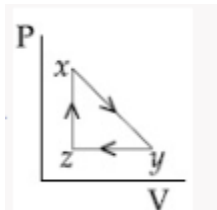
(a)



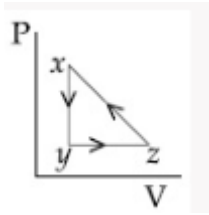
(b)



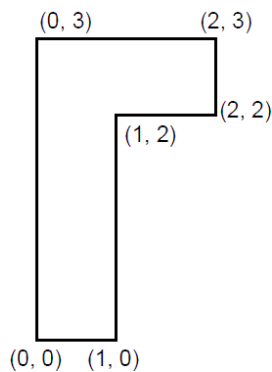
(c)



(d)



Question 19: The coordinates of centre of mass of a uniform flag shaped lamina (thin flat plate) of mass 4 kg. (The coordinates of the same are shown in figure) are



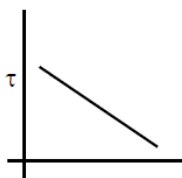
Options:

- (a) (1.25 m, 1.50 m)
- (b) (0.75 m, 0.75 m)
- (c) (0.75 m, 1.75 m)
- (d) (1 m, 1.75 m)

Question 20: The plot that depicts the behaviour of the mean free time (τ) (time between two successive collisions for the molecules of an ideal gas, as a function of temperature (T), qualitatively, is : (Graphs are schematic and not drawn to scale)

Options:

- (a)
- (b)
- (c)
- (d)



Question 21: A body A, mass $m=0.1$ kg has an initial velocity of $3\hat{i}\text{ ms}^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{i}\text{ ms}^{-1}$. The energy of B after collision is written as $x/10$ J. The value of x is _____

Question 22: A particle is moving along the x-axis with its coordinates with time 't' given by $x(t) = 10+8t-3t^2$. Another particle is moving along the y-axis with its coordinate as a function of time given by $y(t) = 5-8t^3$. At $t = 1$ s, the speed of the second particle measured in the frame of the first particle is given as $\sqrt{v}$. The v (in m/s) is _____

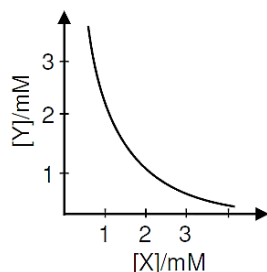
Question 23: A one metre long (both ends open) organ pipe is kept in a gas that has double the density of air at STP. Assuming the speed of sound in air at STP is 300 m/s , the frequency difference between the fundamental and second harmonic of this pipe is _____ Hz.

Question 24: A point object is in front of the curved surface of a plano-convex lens. The radius of curvature of the curved surface is 30 cm and the refractive index of the lens material is 1.5 , then the focal length of the lens (in cm) is _____

Question 25: Four resistances of 15Ω , 12Ω , 4Ω and 10Ω respectively in cyclic order to form Wheatstone's network. The resistance that is to be connected in parallel with the resistance of 10Ω to balance the network is _____ Ω

CHEMISTRY

Question 26: The stoichiometry and solubility product of a salt with the solubility curve given below is, respectively



Options:

- (a) X_2Y , $2 \times 10^{-9}\text{ M}^3$
- (b) XY , $2 \times 10^{-6}\text{ M}^3$
- (c) XY_2 , $4 \times 10^{-9}\text{ M}^3$

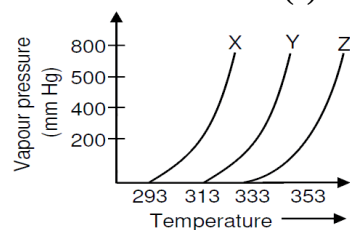
(d) XY_2 , $1 \times 10^{-9} \text{ M}^3$

Question 27: A graph of vapour pressure and temperature for three different liquids X, Y and Z is shown below

The following inferences are made

- (1) X has higher intermolecular interactions compared to Y.
- (2) X has lower intermolecular interactions compared to Y.
- (3) Z has lower intermolecular interactions compared to Y.

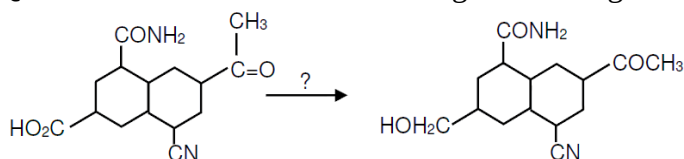
The correct inference(s) is/are



Options:

- (a) (C)
- (b) (B)
- (c) (A) and (C)
- (d) (A)

Question 28: The most suitable reagent for the given conversion is



Options:

- (a) H_2/Pd
- (b) NaBH_4
- (c) LiAlH_4
- (d) B_2H_6

Question 29: The complex that can show fac– and mer– isomers is

Options:

- (a) $[\text{CoCl}_2(\text{en})_2]$
- (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
- (c) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$
- (d) $[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$

Question 30: The rate of a certain biochemical reaction at physiological temperature (T) occurs 10^6 times faster with enzyme than without. The change in the activation energy upon adding enzyme is

Options:

- (a) $+6(2.303)RT$
- (b) $+6RT$

- (c) $-6RT$
 (d) $-6(2.303) RT$

Question 31: The first ionization energy (in kJ/mol) of Na, Mg, Al and Si respectively, are

Options:

- (a) 496, 577, 737, 786
 (b) 786, 737, 577, 496
 (c) 496, 577, 786, 737
 (d) 496, 737, 577, 786

Question 32: The third ionization enthalpy is minimum for

Options:

- (a) Ni
 (b) Co
 (c) Mn
 (d) Fe

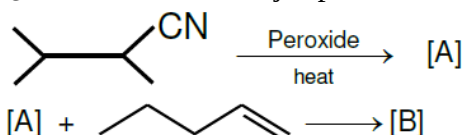
Question 33: Among the gases (a) – (e), the gases that cause greenhouse effect are

- (a) CO_2 (b) H_2O (c) CFCs (d) O_2 (e) O_3

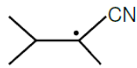
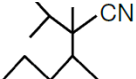
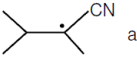
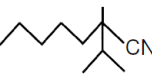
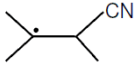
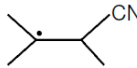
Options:

- (a) (a), (c), (d) and (e)
 (b) (a), (b), (c) and (e)
 (c) (a), (b), (c) and (d)
 (d) (a) and (d)

Question 34: The major products A and B in the following reactions are



Options:

- (a)
 A =  and तथा B = 
 (b)
 A =  and तथा B = 
 (c)
 A =  and तथा B = 
 (d)
 A =  and तथा B = 

Question 35: The predominant intermolecular forces present in ethyl acetate, a liquid, are

Options:

- (a) Dipole-dipole and hydrogen bonding
- (b) London dispersion, dipole-dipole and hydrogen bonding
- (c) London dispersion and dipole-dipole
- (d) Hydrogen bonding and London dispersion

Question 36: Arrange the following compounds in increasing order of C–OH bond length
Methanol, phenol, p-ethoxyphenol

Options:

- (a) methanol < phenol < p-ethoxyphenol
- (b) phenol < methanol < p-ethoxyphenol
- (c) phenol < p-ethoxyphenol < methanol
- (d) methanol < p-ethoxyphenol < phenol

Question 37: A flask contains a mixture of isohexane and 3-methylpentane. One of the liquids boils at 63°C while the other boils at 60°C. What is the best way to separate the two liquids and which one will be distilled out first?

Options:

- (a) simple distillation, isohexane
- (b) simple distillation, 3-methylpentane
- (c) fractional distillation, 3-methylpentane
- (d) fractional distillation, isohexane

Question 38: For the Balmer series in the spectrum of H atom, $\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$ the correct

statements among (I) to (IV) are

- (I) As wavelength decreases, the lines in the series converge
- (II) The integer n_1 is equal to 2
- (III) The lines of longest wavelength correspond to $n_2 = 3$
- (IV) The ionization energy of hydrogen can be calculated from wave number of these lines

Options:

- (a) (I), (III), (IV)
- (b) (I), (II), (IV)
- (c) (II), (III), (IV)
- (d) (I), (II), (III)

Question 39: The number of bonds between sulphur and oxygen atoms in $\text{S}_2\text{O}_8^{2-}$ and the number of bonds between sulphur and sulphur atoms in rhombic sulphur, respectively, are

Options:

- (a) 8 and 8
- (b) 8 and 6
- (c) 4 and 6
- (d) 4 and 8

Question 40: The strength of an aqueous NaOH solution is most accurately determined by titrating (Note consider that an appropriate indicator is used)

Options:

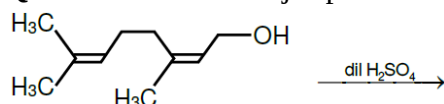
- (a) Aq. NaOH in a volumetric flask and concentrated H_2SO_4 in a conical flask
- (b) Aq. NaOH in a burette and aqueous oxalic acid in a conical flask
- (c) Aq. NaOH in a pipette and aqueous oxalic acid in a burette
- (d) Aq. NaOH in a burette and concentrated H_2SO_4 oxalic in a conical flask

Question 41: As per Hardy–Schulze formulation, the flocculation values of the following for ferric hydroxide sol are in the order

Options:

- (a) $\text{K}_3[\text{Fe}(\text{CN})_6] < \text{K}_2\text{CrO}_4 < \text{KBr} = \text{KNO}_3 = \text{AlCl}_3$
- (b) $\text{K}_3[\text{Fe}(\text{CN})_6] < \text{K}_2\text{CrO}_4 < \text{AlCl}_3 < \text{KBr} < \text{KNO}_3$
- (c) $\text{K}_3[\text{Fe}(\text{CN})_6] > \text{AlCl}_3 > \text{K}_2\text{CrO}_4 > \text{KBr} > \text{KNO}_3$
- (d) $\text{AlCl}_3 > \text{K}_3[\text{Fe}(\text{CN})_6] > \text{K}_2\text{CrO}_4 > \text{KBr} = \text{KNO}_3$

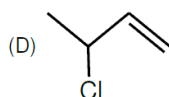
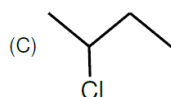
Question 42: The major product of the following reaction is



Options:

- (a)
- (b)
- (c)
- (d)

Question 43: The decreasing order of reactivity towards dehydrohalogenation (E1) reaction of the following compounds is



Options:

- (a) $B > D > C > A$
- (b) $D > B > C > A$
- (c) $B > A > D > C$
- (d) $B > D > A > C$

Question 44: When gypsum is heated to 393 K, it forms

Options:

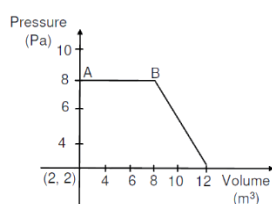
- (a) Anhydrous CaSO_4
- (b) $\text{CaSO}_4 \cdot 0.5 \text{H}_2\text{O}$
- (c) $\text{CaSO}_4 \cdot 5 \text{H}_2\text{O}$
- (d) Dead burnt plaster

Question 45: Which of the following statement is not true for glucose

Options:

- (a) Glucose exists in two crystalline forms α and β
- (b) Glucose reacts with hydroxylamine to form oxime
- (c) Glucose gives Schiff's test for aldehyde
- (d) The pentaacetate of glucose does not react with hydroxylamine to give oxime

Question 46: The magnitude of work done by a gas that undergoes a reversible expansion along the path ABC shown in the figure is _____



Question 47: The volume (in mL) of 0.125 M AgNO_3 required to quantitatively precipitate chloride ions in 0.3 g of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is _____.

$$M[\text{Co}(\text{NH}_3)_6]\text{Cl}_3 = 267.46 \text{ g/mol}$$

$$M\text{AgNO}_3 = 169.87 \text{ g/mol}$$

Question 48: Ferrous sulphate heptahydrate is used to fortify foods with iron. The amount (in grams) of the salt required to achieve 10 ppm of iron in 100 kg of wheat is _____.

$$\text{Atomic weight Fe} = 55.85 \quad \text{S} = 32.00 \quad \text{O} = 16.00$$

Question 49: What would be the electrode potential for the given half-cell reaction at pH = 5
 $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4\text{e}^-$ $E_0^{\text{red}} = 1.23 \text{ V}$ ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ Temp = 298 K oxygen under std. pressure of 1 bar)

Question 50: The number of chiral centres in penicillin is

MATHEMATICS

Question 51: $\lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}}$ is equal to

Options:

- (a) $1/e^2$
- (b) e
- (c) e^2
- (d) $1/e$

Question 52: Let two points be A(1, -1) and B(0, 2). If a point P(x', y') be such that the area of $\Delta \text{PAB} = 5$ square. units and it lies on the line, $3x + y - 4\lambda = 0$, then a value of λ is :

Options:

- (a) 3
- (b) -3
- (c) 1
- (d) 4

Question 53: The mean and the standard deviation (s, d) of 10 observations are 20 and 2 respectively. Each of these 10 observations is multiplied by p and then reduced by q, where $p \neq 0$ and $q \neq 0$. If the new mean and new s.d. become half of their original values, then q is equal to :

Options:

- (a) -20
- (b) 10
- (c) -10
- (d) -5

Question 54: The shortest distance between the lines

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-8}{1} \text{ and } \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} \text{ is}$$

Options:

- (a) $2\sqrt{30}$

- (b) 3
 (c) $\frac{7}{2}\sqrt{30}$
 (d) $3\sqrt{30}$

Question 55: If a, b and c are the greatest values of P19C , Q20C and R21C respectively, then :

Options:

- (a) $\frac{a}{10} = \frac{b}{11} = \frac{c}{42}$
 (b) $\frac{a}{10} = \frac{b}{11} = \frac{c}{21}$
 (c) $\frac{a}{11} = \frac{b}{22} = \frac{c}{42}$
 (d) $\frac{a}{11} = \frac{b}{22} = \frac{c}{21}$

Question 56: Let the volume of a parallelopiped whose coterminous edges are given by $\vec{u} = \hat{i} + \hat{j} + \lambda\hat{k}$, $\vec{v} = \hat{i} + \hat{j} + 3\hat{k}$ and $\vec{w} = 2\hat{i} + \hat{j} + \hat{k}$ be 1 cu. unit. If θ be the angle between the edges $\vec{u}$ and $\vec{w}$ then $\cos\theta$ can be

Options:

- (a) $\frac{5}{3\sqrt{3}}$
 (b) $\frac{7}{6\sqrt{6}}$
 (c) $\frac{5}{7}$
 (d) $\frac{7}{6\sqrt{3}}$

Question 57: Let $f(x) = x\cos^{-1}(-\sin|x|)$, $x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, then which of the below is true ?

Options:

- (a) $f'(0) = -\frac{\pi}{2}$
 (b) f' is decreasing in $\left(-\frac{\pi}{2}, 0\right)$ and increasing in $\left(0, \frac{\pi}{2}\right)$
 (c) f is not differentiable at $x = 0$
 (d) f' is increasing in $\left(-\frac{\pi}{2}, 0\right)$ and decreasing in $\left(0, \frac{\pi}{2}\right)$

Question 58: Let $f(x) = (\sin(\tan^{-1}x) + \sin(\cot^{-1}x))^2 - 1$, $|x| > 1$. If $\frac{dy}{dx} = \frac{1}{2} \frac{d}{dx}(\sin^{-1}(f(x)))$

and $y(\sqrt{3}) = \frac{\pi}{6}$, then $y(-\sqrt{3})$ is equal to :

Options:

- (a) $\pi/3$
- (b) $2\pi/3$
- (c) $-\pi/6$
- (d) $5\pi/6$

Question 59: Let A and B be two independent events such that $P(A) = \frac{1}{3}$ and

$P(B) = \frac{1}{6}$. Then, which of the following is TRUE?

Options:

- (a) $P(A/B) = \frac{1}{3}$
- (b) $P(A/B) = \frac{2}{3}$
- (c) $P(A/B) = \frac{1}{3}$
- (d) $P(A/(A \cup B)) = \frac{1}{4}$

Question 60: The inverse function of $f(x) = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}}$, $x \in (-1, 1)$, is

Options:

- (a) $\frac{1}{4} \log_e \left(\frac{1+x}{1-x} \right)$
- (b) $\frac{1}{4} \log_e \left(\frac{1-x}{1+x} \right)$
- (c) $\frac{1}{4} (\log_e e) \left(\frac{1+x}{1-x} \right)$
- (d) $\frac{1}{4} (\log_e e) \left(\frac{1-x}{1+x} \right)$

Question 61: If c is a point at which Rolle's theorem holds for the function

$f(x) = \log_e \left(\frac{x^2 + \alpha}{7x} \right)$ in the interval $[3, 4]$, where $\alpha \in \mathbb{R}$, then $f'(c)$ is equal to :

Options:

- (a) $\frac{\sqrt{3}}{7}$
 (b) $-\frac{1}{24}$
 (c) $-\frac{1}{12}$
 (d) $\frac{1}{12}$

Question 62: Let $y = y(x)$ be a solution of the differential equation,

$$\sqrt{1-x^2} \frac{dy}{dx} + \sqrt{1-y^2} = 0, |x| < 1.$$

If $y\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$, then $y\left(\frac{-1}{\sqrt{2}}\right)$ is equal to

- (a) $-\frac{\sqrt{3}}{2}$
 (b) $\frac{1}{\sqrt{2}}$
 (c) $\frac{\sqrt{3}}{2}$
 (d) $-\frac{1}{\sqrt{2}}$

Question 63: If $\int \frac{\cos x \, dx}{\sin^3 x (1 + \sin^6 x)^{2/3}} = f(x) (1 + \sin^6 x)^{1/\lambda} + c$ where c is a constant of

integration, then $\lambda f\left(\frac{\pi}{3}\right)$ is equal to :

Options:

- (a) 2
 (b) $-\frac{9}{8}$
 (c) $\frac{9}{8}$
 (d) -2

Question 64: Let the line $y = mx$ and the ellipse $2x^2 + y^2 = 1$ intersect at a point P in the first quadrant. If the normal to this ellipse at P meets the co-ordinate axes at $\left(-\frac{1}{3\sqrt{2}}, 0\right)$ and

$(0, \beta)$, then β is equal to :

Options:

- (a) $\frac{2}{\sqrt{3}}$
- (b) $\frac{\sqrt{2}}{3}$
- (c) $\frac{2}{3}$
- (d) $\frac{2\sqrt{2}}{3}$

Question 65: The locus of a point which divides the line segment joining the point $(0, -1)$ and a point on the parabola, $x^2 = 4y$, internally in the ratio $1 : 2$, is :

Options:

- (a) $9x^2 - 12y = 8$
- (b) $x^2 - 3y = 2$
- (c) $9x^2 - 3y = 2$
- (d) $4x^2 - 3y = 2$

Question 66: Which one of the following is a tautology?

Options:

- (a) $(P \wedge (P \rightarrow Q)) \rightarrow Q$
- (b) $(P \wedge (P \vee Q))$
- (c) $Q \rightarrow (P \wedge (P \rightarrow Q))$
- (d) $(P \vee (P \wedge Q))$

Question 67: For $a > 0$, let the curves $C_1: y^2 = ax$ and $C_2: x^2 = ay$ intersect at origin O and a point P. Let the line $x = b$ ($0 < b < a$) intersect the chord OP and the x-axis at points Q and R, respectively. If the line $x = b$ bisects the area bounded by the curves, C_1 and C_2 , and the area of $\Delta OQR = \frac{1}{2}$, then 'a' satisfies the equation :

Options:

- (a) $x^6 - 12x^3 - 4 = 0$
- (b) $x^6 - 6x^3 + 4 = 0$
- (c) $x^6 - 12x^3 + 4 = 0$
- (d) $x^6 + 6x^3 - 4 = 0$

Question 68: If the equation, $x^2 + bx + 45 = 0$ ($b \in \mathbb{R}$) has conjugate complex roots and they satisfy $|z+1| = 2\sqrt{10}$, then :

Options:

- (a) $b^2 - b = 30$
- (b) $b^2 + b = 72$
- (c) $b^2 - b = 42$

(d) $b^2 + b = 12$

Question 69: Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that for all $x \in \mathbb{R}$ ($2^{1+x} + 2^{1-x}$), $f(x)$ and $(3^x + 3^{-x})$ are in A.P., then the minimum value of $f(x)$ is :

Options:

- (a) 0
- (b) 1
- (c) 4
- (d) 3

Question 70: For which of the following ordered pairs (μ, δ) , the system of linear equations
 $x + 2y + 3z = 1$
 $3x + 4y + 5z = \mu$
 $4x + 4y + 4z = \delta$ is inconsistent ?

Options:

- (a) (4, 6)
- (b) (3, 4)
- (c) (1, 0)
- (d) (4, 3)

Question 71: The number of all 3×3 matrices A , with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3, is _____

Question 72: The least positive value of 'a' for which the equation,

$2x^2 + (a - 10)x + \frac{33}{2} = 2a$ has real roots is

Question 73: An urn contains 5 red marbles, 4 black marbles and 3 white marbles. Then the number of ways in which 4 marbles can be drawn so that at the most three of them are red is _____.

Question 74: The sum $\sum_{k=1}^{20} (1 + 2 + 3 + \dots + k)$ is

Question 75: Let the normal at a point P on the curve $y^2 - 3x^2 + y + 10 = 0$ intersect the y -axis at $(0, 3/2)$. If m is the slope of the tangent at P to the curve, then $|m|$ is equal to

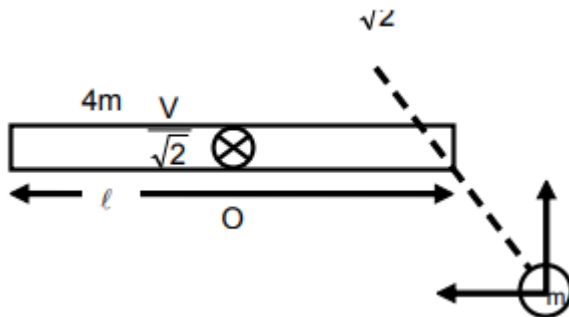
ANSWER KEY

1. (a)	2. (d)	3. (c)	4. (a)	5. (d)	6. (b)	7. (d)	8. (b)	9. (d)	10. (a)
11. (d)	12. (Bonus)	13. (b)	14. (a)	15. (c)	16. (a)	17. (d)	18. (b)	19. (c)	20. (c)
21. (60.00)	22. (106.00)	23. (580.00)	24. (1.00)	25. (10.00)	26. (d)	27. (b)	28. (d)	29. (d)	30. (d)
31. (c)	32. (d)	33. (b)	34. (b)	35. (c)	36. (c)	37. (d)	38. (d)	39. (a)	40. (b)
41. (a)	42. (c)	43. (b)	44. (b)	45. (c)	46. (1.52)	47. (26. 80)	48. (4. 95)	49. (48.00)	50. (3. 00)
51. (a)	52. (b)	53. (a)	54. (d)	55. (c)	56. (d)	57. (b)	58. (d)	59. (c)	60. (c)
61. (d)	62. (b)	63. (d)	64. (b)	65. (a)	66. (a)	67. (c)	68. (a)	69. (d)	70. (d)
71. (672.00)	72. (8.00)	73. (490.00)	74. (1540.00)	75. (4.00)					

SOLUTIONS:

PHYSICS

Solution 1:



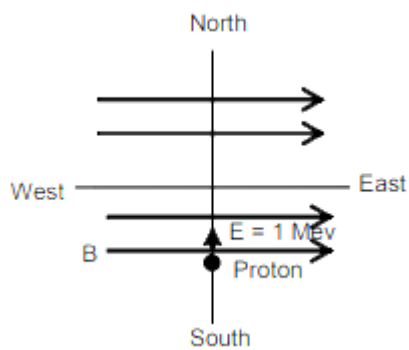
$$L_{Oi} = L_{Of}$$

$$\omega = \frac{6v}{7\sqrt{2}l} = \frac{3\sqrt{2}v}{7l}$$

Solution 2:

$$\begin{aligned} \varepsilon &= \left| -\frac{d\phi}{dt} \right| = \left| -\frac{AdB}{dt} \right| \\ &= 56 \times \frac{500}{5} \times 10^{-8} = 56 \times 10^{-6} \text{ V} \end{aligned}$$

Solution 3:



$$\therefore \text{K.E} = 1.6 \times 10^{-13} = \frac{1}{2} \times 1.6 \times 10^{-27} v^2$$

$$v = \sqrt{2} \times 10^7 \text{ m/s}$$

$$\therefore Bqv = ma$$

$$B = \frac{1.6 \times 10^{-27} \times 10^{12}}{1.6 \times 10^{-19} \times \sqrt{2} \times 10^7}$$

So, 0.71 mT

Solution 4:

Magnitude of electric field is constant & the surface is equipotential

Solution 5:

Given $C_1 + C_2 = 10\mu\text{F} \dots (1)$

$$4\left(\frac{1}{2}C_1V^2\right) = \frac{1}{2}C_2V^2$$

From equation (1) and (2)

$$C_1 = 2\mu\text{F}$$

$$C_2 = 8\mu\text{F}$$

In series

$$C_{\text{eq}} = \frac{C_1C_2}{C_1 + C_2} = 1.6\mu\text{F}$$

Solution 6:

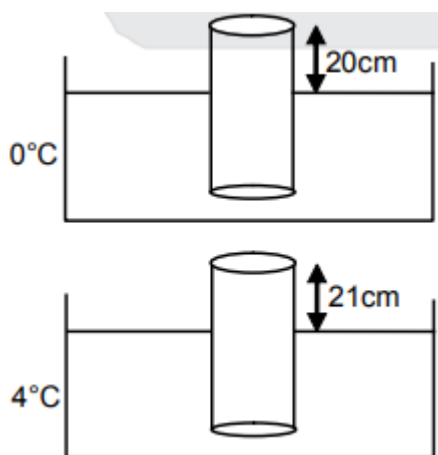
$$3 = \frac{Gm_2}{2^2}$$

$$2 = \frac{Gm_1}{1^2}$$

$$\frac{m_1}{m_2} = \frac{1}{6}$$

Solution 7:

The density of water at $T = 4^\circ\text{C}$, relative to the density at $T = 0^\circ\text{C}$ is close to



$$mg = A(80)\rho^{0^\circ\text{C}}g$$

$$mg = A(79)\rho^{4^\circ\text{C}}g$$

$$\frac{\rho^{4^\circ\text{C}}}{\rho^{0^\circ\text{C}}} = \frac{80}{79} = 1.01$$

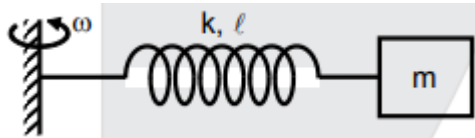
Solution 8:

$$m = \frac{f_0}{f_e}$$

$$f_0 = 5f_e$$

$$f_0 + f_e = 60$$

$$f_e = 10$$

Solution 9:

$$m\omega^2(l+x) = kx$$

$$x = \frac{lm\omega^2}{k - m\omega^2}$$

Solution 10:

Rutherford gold foil experiment with α -particles

$$N \propto \frac{1}{\sin^4\left(\frac{\theta}{2}\right)}$$

Solution 11:

First part of figure shown is OR gate and Second part of figure shown is NOT gate NOT

$$Y_p = OR + NOT = NOR$$

$$Y = \overline{A + B} = \overline{A} \cdot \overline{B}$$

Solution 12:

$$\lambda = \frac{h}{\sqrt{2(KE)m_e}} \Rightarrow \lambda \propto \frac{1}{\sqrt{KE}}$$

$$\therefore KE_B = 2 - 1.5 = 0.5 \text{ eV}$$

$$\phi_B = 4.5 - 0.5 = 4 \text{ eV}$$

Solution 13:

$$V = \frac{1}{\sqrt{\mu\epsilon}}$$

$$\sin c = \frac{1}{2}$$

$$c = 30^\circ$$

Solution 14:

$$\rho = \rho_0 \left(1 - \frac{r^2}{R^2} \right) \quad 0 < r \leq R$$

$$mg = B$$

$$\int_0^R \rho_0 4\pi \left(r^2 - \frac{r^4}{R^2} \right) dr = \rho_0 4\pi \left(\frac{r^3}{3} - \frac{r^5}{5R^2} \right) \Big|_0^R = \rho L \frac{4}{3} \pi R^3, \quad \frac{2}{5} \rho_0 = \rho L$$

Solution 15:

$$E_{\text{net}} = \frac{4kq}{d^2} \times 2 \cos 30^\circ = \frac{q\sqrt{3}}{\pi \epsilon_0 d^2}$$

Solution 16:

$$V = K (h)^a (I)^b (G)^c (C)^d$$

$$M L^2 T^{-3} A^{-1} = (M L^2 T^{-1})^a (A)^b (M^{-1} L^3 T^{-2})^c (L T^{-1})^d$$

$$M L^2 T^{-3} A^{-1} = M^{a-c} L^{2a+3c+d} T^{-a-2c-d} A^b$$

$$a - c = 1 \quad \dots\dots\dots(1)$$

$$2a + 3c + d = 2 \quad \dots\dots\dots(2)$$

$$-a - 2c - d = -3 \quad \dots\dots\dots(3)$$

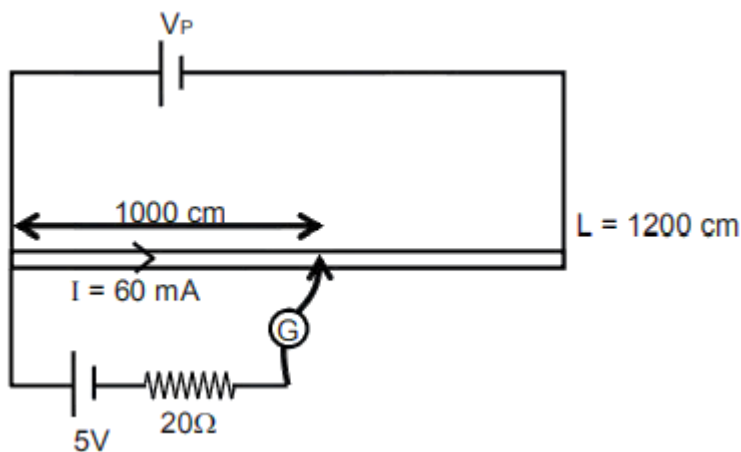
$$b = -1 \quad \dots\dots\dots(4)$$

$$c = -1$$

$$a = 0$$

$$d = 5, b = -1$$

$$V = K (h)^0 (I)^{-1} (G)^{-1} (C)^5$$

Solution 17:

$$\text{Potential gradient} = \frac{5}{1000} = \frac{V_p}{1200}$$

$$V_p = 6V$$

$$R_p = \frac{V_p}{I} = \frac{6}{60 \times 10^{-3}} = 100 \Omega$$

Solution 19:

$$\vec{r}_{\text{cm}} = \frac{1 \times \left(\frac{\hat{i}}{2} + \hat{j} \right) + 1 \times \left(\hat{i} + \frac{5\hat{j}}{2} \right)}{2}$$

$$\vec{r}_{\text{cm}} = \frac{3}{4}\hat{i} + \frac{7}{4}\hat{j}$$

Solution 20:

The plot that depicts the behaviour of the mean free time (τ) (time between two successive collisions for the molecules of an ideal gas

$$\tau \propto \frac{1}{\sqrt{T}}$$

Solution 21:

$$\text{KE}_i = \text{KE}_f$$

$$34 = 32 + v^2$$

$$\text{KE} = \frac{1}{2} \times 0.1 \times 2 = 0.1 \text{ J} = \frac{1}{10}$$

$$x = 1$$

Solution 22:

$$X_1 = -3t^2 + 8t + 10$$

$$\vec{v}_1 = (-6t + 8)\hat{i} = 2\hat{i}$$

$$Y_2 = 5 - 8t^3$$

$$\vec{v}_2 = -24t^2\hat{j}$$

$$\sqrt{v} = |\vec{v}_2 - \vec{v}_1| = |-24\hat{j} - 2\hat{i}|$$

$$v = 580$$

Solution 23:

$$V = \sqrt{\frac{B}{\rho}}$$

$$\frac{V_{\text{pipe}}}{V_{\text{air}}} = \frac{\sqrt{\frac{B}{2\rho}}}{\sqrt{\frac{B}{\rho}}} = \frac{1}{\sqrt{2}}$$

$$V_{\text{pipe}} = \frac{V_{\text{air}}}{\sqrt{2}}$$

$$f_n = \frac{(n+1)V_{\text{pipe}}}{2l}$$

$$f_1 - f_0 = \frac{V_{\text{pipe}}}{2l} = \frac{300}{2\sqrt{2}}$$

$$= 106.05 \text{ Hz (If } \sqrt{2} = 1.414 \text{)}$$

Solution 24:

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

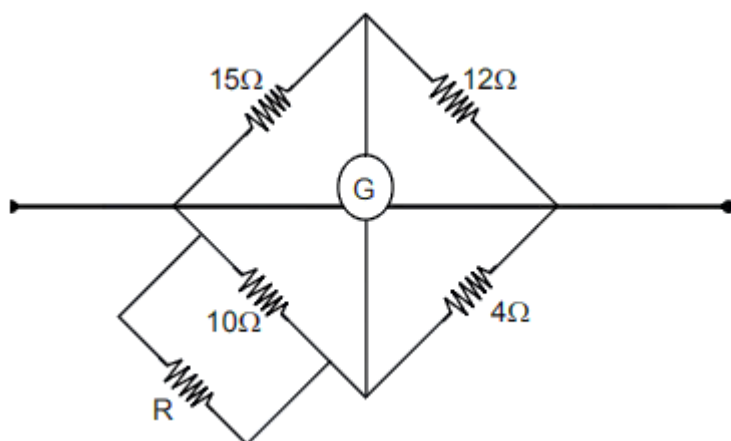
$$R_1 = \infty$$

$$R_2 = -30 \text{ cm.}$$

$$\frac{1}{f} = (1.5 - 1) \left(\frac{1}{\infty} - \frac{1}{-30} \right)$$

$$f = 60 \text{ cm}$$

Solution 25:

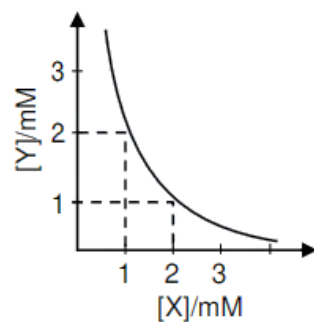


$$\frac{10R}{10 + R} \times 12 = 15 \times 4$$

$$R = 10\Omega$$

CHEMISTRY

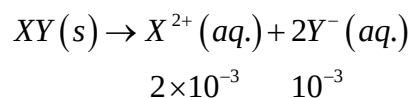
Solution 26:



From the graph and dimension salt is XY_2

$$[X] = 1 \times 10^{-3}$$

$$[Y] = 2 \times 10^{-3}$$



$$K_{sp} = [X^{2+}] \cdot [Y^{-}]^2$$

$$= (2 \times 10^{-3})^2 \times 10^{-3}$$

$$= 4 \times 10^{-9}$$

Corresponding stoichiometry = XY_2

$$K_{sp} = 4 \times 10^{-9} M^3$$

Solution 27: 'X' has lower intermolecular interactions as compared to 'Y'. Because at a particular temperature as the intermolecular force of attraction increases vapour pressure decreases. At a particular temperature as intermolecular force of attraction increases vapour pressure decreases.

Solution 28: B_2H_6 is very selective reducing agent it is usually used to reduce acid to alcohol

Solution 29:

$[Ma_3b_3]$ type complex shows facial and meridional isomerism



fac isomer



mer isomer

Solution 30:

$$\text{Rate of reaction (K)} = Ae^{-E/RT} \quad \dots \dots (i)$$

$$\text{and, } 10^6 \cdot K = Ae^{-E_c/RT} \quad \dots \dots (ii)$$

by dividing equation (ii) by (i):

$$\frac{10^6 \cdot K}{K} = \frac{Ae^{-E_c/RT}}{Ae^{-E/RT}}$$

$$10^6 = e^{(E-E_c)/RT}$$

$$\Rightarrow 6 \cdot \ln 10 = (E - E_c) / RT$$

$$\Rightarrow \frac{E - E_c}{RT} = 2.303 \times 6$$

$$\Rightarrow E - E_c = 6 \times 2.303RT$$

$$\text{or } \Delta E_a = E_c - E = -6 \times 2.303RT$$

Solution 31: Sodium has one electron in valence shell Magnesium has ns^2 so stable configuration

Correct order of ionisation energy will be $Na < Al < Mg < Si$

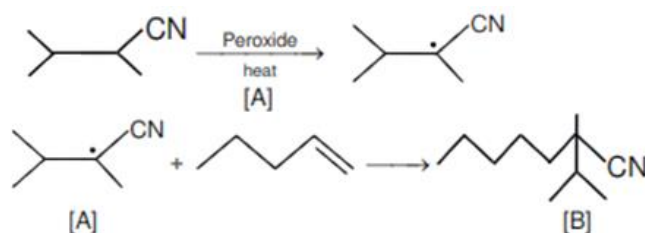
Solution 32: after losing three electrons iron will gain stable half-filled configuration. So, it will have least 3rd ionization energy



Solution 33:

Carbon dioxide, water vapours chloroflouro carbon and ozone gas are greenhouse gases.

Solution 34:



[A] would be more stable radical and undergoes radical addition to form [B].

Solution 35: The predominant intermolecular forces present in ethyl acetate liquid is London dispersion and dipole-dipole interaction. Ethyl acetate is a polar molecule. Therefore, dipole-dipole interaction will be present there

Solution 36: glucose does not react with Schiff's reagent although it has an aldehydic group. After the internal cyclisation, it forms α -anomer. In these forms free aldehydic group is **not** present. So, it does not give a reaction of aldehydic group.

Solution 37:

In $\text{CH}_3\text{-OH}$, there is no resonance. In p-Ethoxyphenol there is lesser resonance than phenol. Due to resonance single bond will have partial double bond character. So, C–OH bond length order is: $\text{CH}_3\text{-OH} > \text{p-ethoxyphenol (p-EtO-C}_6\text{H}_4\text{-OH)} > \text{Ph-OH}$

Solution 38:

$$\bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

(I) As the wavelength decreases the energy increases therefore the spectrum lines become bond and formed a converging series.

(II) For balmer series always started ground state $n_1 = 2$

(III) At longest wavelength, the higher state energy will be minimum therefore excited state will be $n_2 - 3$ next to the ground always.

(IV) $E = -13.6 \frac{z^2}{n^2}$ for H atom $z = 1$

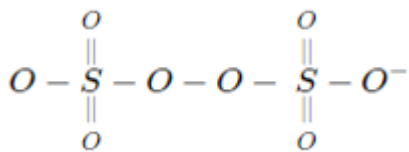
So, we cannot find ionization energy directly by only wave number, we also required transition state(n) value.

According to the above observation

Only (I), (II) and (III) correct.

Solution 39:

In $S_2O_8^{2-}$,



No. of S – O bond = 8

Rhombic sulphur (S_8)

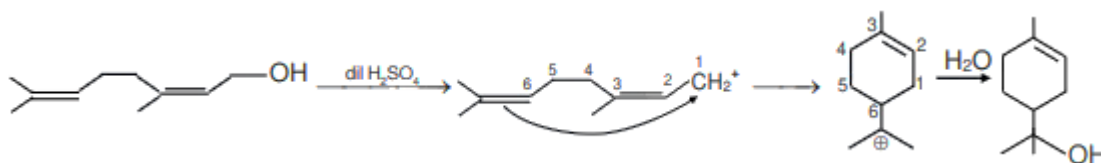
No of S – S bond = 8

Solution 40:

Oxalic acid being primary standard solution and H_2SO_4 being secondary standard solution, oxalic acid will be used for calculation of strength of NaOH.

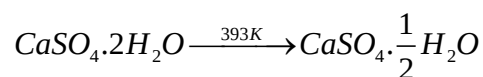
Solution 41: According to hardy-schulze rule, Greater is the valency of the oppositely charged ion of the electrolyte being added, the faster is the cogulation.

$$\text{Coagulation value or flocculation value} \propto \frac{1}{\text{Coagulation power}}$$

Solution 42:

Solution 43: in elimination reaction of first order, reaction proceeds via carbocation formation, therefore the stability of carbocation decides the speed of E1 reaction. Greater stability of carbocation faster the reaction will be

Solution 44: When gypsum is heated at 373 K, it loses its water of crystallization or all the water molecules and becomes calcium sulphate hemihydrate



Solution 45: Fractional distillation is generally used if boiling point difference is small. (Boiling point of 3-methylpentane = $63^\circ C$, isohexane = $60^\circ C$) Liquid having lower boiling point comes out first in fractional distillation

Solution 46:

From the given data

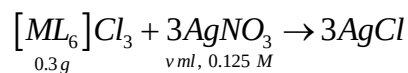
$$E_{OP} = E^\circ_{OP} - \frac{0.059}{4} \log [H^+]^4$$

$$E_{OP} = -1.23 - \frac{0.0591}{4} \log [H^+]^4$$

$$= -1.23 + 0.0591 \times \text{pH} = -1.23 + 0.0591 \times 5$$

$$= -1.23 + 0.2955 = -0.9345 \text{ V} = -0.93 \text{ V}$$

Solution 47:



$$\frac{0.3}{267.46} \times 3 = 0.125 \times V \times 10^{-3}$$

$$\text{or } V = \frac{0.3 \times 3 \times 1000}{267.46 \times 0.125} = 26.92 \text{ ml}$$

Solution 48:

$$10 = \frac{\text{Mass of Fe (ing)}}{100 \times 1000} \times 10^6$$

Mass of Fe = 1 gm

Molar mass of $FeSO_4 \cdot 7H_2O = 277.85$

Mass of Fe is 56 gm in 1 mole

So, 1 gm in $\frac{1}{56}$ mole

$$\text{Therefore, } \frac{1}{56} \times 277.85 = 4.97$$

Solution 49:

Work done = Area covered by the diagram

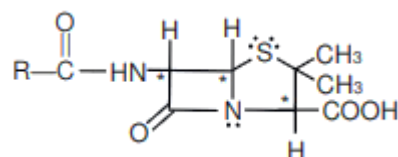
$$= \frac{1}{2} \times (\text{sum of parallel sides}) \times \text{height}$$

$$= \frac{1}{2} \times (10 + 6) \times 6$$

$$= \frac{1}{2} \times 16 \times 6$$

$$= 48 \text{ J}$$

Solution 50:



Star marked atoms are chiral centres. Penicillin has three chiral centres.

MATHEMATICS

Solution 51:

$$L = \lim_{x \rightarrow 0} \left(\frac{3x^2 + 2}{7x^2 + 2} \right)^{\frac{1}{x^2}} = e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{3x^2 + 2}{7x^2 + 2} - 1 \right\}} = e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left\{ \frac{-4x^2}{7x^2 + 2} \right\}} = e^{-\frac{4}{2}} = e^{-2}$$

Solution 52:

$$\text{Consider } D = \frac{1}{2} \begin{vmatrix} 0 & 2 & 1 \\ 1 & -1 & 1 \\ x' & y' & 1 \end{vmatrix}$$

This gives us $-2(1 - x') + (y' + x') = \pm 10$

That is, $-2 + 2x' + y' = \pm 10$

So, we have $3x' + y' = 12$ or $3x' + y' = -8$ and hence, $\lambda = 3, -2$

Solution 53:

Each datum is multiplied by p and then the value q is subtracted from each datum.

Thus, the new mean $\bar{x}_1 = p\bar{x} - q \Rightarrow 10 = p(20) - q \quad \dots(1)$

And the new standard deviation is as follows.

$$\sigma = |p|\sigma_1 \Rightarrow 1 = |p|(2) \Rightarrow |p| = \frac{1}{2} \Rightarrow p = \pm \frac{1}{2}$$

For $p = \frac{1}{2}$, we get $q = 0$ (from equation (1))

For $p = -\frac{1}{2}$, we get $q = -20$

Solution 54:

$$\bar{w} = \hat{i} + \hat{j} - 3\hat{k}, \quad \bar{p} = 3\hat{i} - \hat{j} + \hat{k} \quad \text{and} \quad \bar{q} = -3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\text{So, } \bar{p} \times \bar{q} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -1 & 1 \\ -3 & 2 & 4 \end{vmatrix} = -6\hat{i} - 15\hat{j} + 3\hat{k}$$

$$\text{So, shortest distance is given by } SD = \frac{|\overline{AB} \cdot (\bar{p} \times \bar{q})|}{|\bar{p} \times \bar{q}|} = \frac{|36 + 225 + 9|}{\sqrt{36 + 225 + 9}} = 3\sqrt{30}$$

Solution 55:

We know that the value of nC_r is maximum at the middle term(s)

$$a = {}^{19}C_p = {}^{19}C_{10} = {}^{19}C_9, \quad b = {}^{20}C_q = {}^{20}C_{10} \quad \text{and} \quad c = {}^{21}C_6 = {}^{21}C_{10} = {}^{21}C_{11}$$

$$\text{So, } \frac{a}{{}^{19}C_9} = \frac{b}{\frac{20}{10} \times {}^{19}C_9} = \frac{c}{\frac{21}{11} \times \frac{20}{10} \times {}^{19}C_9}$$

$$\text{This gives us } \frac{a}{1} = \frac{b}{2} = \frac{c}{\frac{42}{11}} \quad \text{or} \quad \frac{a}{11} = \frac{b}{22} = \frac{c}{42}$$

Solution 56:

$$\text{Consider } \pm 1 = \begin{vmatrix} 1 & 1 & \lambda \\ 1 & 1 & 3 \\ 2 & 1 & 1 \end{vmatrix} \Rightarrow -\lambda + 3 = \pm 1 \Rightarrow \lambda = 2 \text{ or } \lambda = 4$$

$$\text{For } \lambda = 4, \text{ the value of } \cos \theta = \frac{2+1+4}{\sqrt{6}\sqrt{18}} = \frac{7}{6\sqrt{3}}$$

Solution 57:

$$\text{We have } f'(x) = x(\pi - \cos^{-1}(\sin|x|)) = \left(\pi - \left(\frac{\pi}{2} - \sin^{-1}(\sin|x|) \right) \right) = x \left(\frac{\pi}{2} + |x| \right)$$

$$\text{So, the function is defined as } f(x) = \begin{cases} x \left(\frac{\pi}{2} + x \right) & x \geq 0 \\ x \left(\frac{\pi}{2} - x \right) & x < 0 \end{cases}$$

$$\text{And its derivative is defined as } f'(x) = \begin{cases} \frac{\pi}{2} + 2x & x \geq 0 \\ \frac{\pi}{2} - 2x & x < 0 \end{cases}$$

Clearly, $f'(x)$ is increasing on the interval $\left(0, \frac{\pi}{2}\right)$ and decreasing on the interval $\left(-\frac{\pi}{2}, 0\right)$

Solution 58:

$$\text{Let } \tan^{-1} x = \theta \Rightarrow x = \tan \theta \Rightarrow \sin \theta = \frac{x}{\sqrt{1+x^2}}$$

$$\text{Now, } y = \left(\frac{x}{\sqrt{1+x^2}} + \frac{1}{\sqrt{1+x^2}} \right)^2 - 1 \Rightarrow y = \frac{(x+1)^2}{1+x^2} - 1$$

$$\text{So, } y = \frac{2x}{1+x^2} = f(x)$$

Now,

$$\frac{dy}{dx} = \frac{1}{2} \frac{1}{\sqrt{1-f^2}} \times f' = \frac{1}{2} \frac{1}{\sqrt{1-\frac{4x^2}{(1+x^2)^2}}} f'(x) = \frac{(1+x^2)}{2(x^2-1)} f'(x) = \frac{1+x^2}{2(x^2-1)} \times 2 \frac{1+x^2-2x^2}{(1+x^2)^2}$$

$$\therefore \int dy = \int \frac{1-x^2}{(x^2-1)(1+x^2)} dx$$

which gives us $y = \tan^{-1} x + c$

$$\text{So, } y(\sqrt{3}) = \frac{\pi}{6} \Rightarrow \frac{\pi}{6} = -\frac{\pi}{3} + c \Rightarrow c = \frac{\pi}{2}$$

Thus, the particular solution is $y = -\tan^{-1} x + \frac{\pi}{2} = \cot^{-1} x$

$$\text{Hence, } y(-\sqrt{3}) = \cot^{-1} x(-\sqrt{3}) = \frac{5\pi}{6}$$

Solution 59:

Since A and B are independent events, $P\left(\frac{A}{B'}\right) = \frac{1}{3}$

Solution 60:

$$y = \frac{8^{2x} - 8^{-2x}}{8^{2x} + 8^{-2x}} \Rightarrow \frac{1+y}{1-y} = \frac{8^{2x}}{8^{-2x}} \Rightarrow 8^{4x} = \frac{1+y}{1-y} \Rightarrow 4x = \log_8 \left(\frac{1+y}{1-y} \right) \Rightarrow$$

$$\therefore f^{-1}(x) = \frac{1}{4} \log_8 \left(\frac{1+x}{1-x} \right)$$

Solution 61:

$$f(3) = f(4) \Rightarrow \alpha = 12$$

$$\text{Now, } f'(x) = \frac{x^2 - 12}{x(x^2 + 12)}$$

$$f'(c) = 0 \Rightarrow c = \sqrt{12} \text{ and } f''(c) = \frac{1}{12}$$

Solution 62:

Solving the differential equation $\frac{dy}{\sqrt{1-y^2}} + \frac{dx}{\sqrt{1-x^2}} = 0$, we get $\sin^{-1} y + \sin^{-1} x = c$.

Now, at $x = \frac{1}{2}$, we have $y = \frac{\sqrt{3}}{2}$.

Substituting these values, we get the value of c as $c = \frac{\pi}{2}$

So, the particular solution is $\sin^{-1} y = \cos^{-1} x$

Thus, $y\left(-\frac{1}{\sqrt{2}}\right) = \sin\left(\cos^{-1}\frac{-1}{2}\right) = \frac{1}{\sqrt{2}}$

Solution 63:

Let $\sin x = t$, so we get $\cos x dx = dt$

Thus, $I = \int \frac{dt}{t^3(1+t^6)^{\frac{2}{3}}} = \int \frac{dt}{t^7\left(1+\frac{1}{t^6}\right)^{\frac{2}{3}}}$

Substituting $1 + \frac{1}{t^6} = r^3$, we get $\frac{dt}{t^7} = -\frac{1}{2}r^2 dr$

Hence, $I = -\frac{1}{2} \int \frac{r^2 dr}{r^2} = -\frac{1}{2}r + c = -\frac{1}{2}\left(\frac{\sin^6 x + 1}{\sin^6}\right)^{\frac{1}{3}} + c = -\frac{1}{2\sin^2 x}(1 + \sin^6 x)^{\frac{1}{3}} + c$

Clearly, $f(x) = -\frac{1}{2}\operatorname{cosec}^2 x$ and $\lambda = 3$, so, $\lambda f\left(\frac{\pi}{3}\right) = -2$

Solution 64:

Let $P \equiv (x_1, y_1)$

The equation of the normal at P is given by $\frac{x}{2x_1} - \frac{y}{y_1} = -\frac{1}{2}$

Since it passes through the point $\left(-\frac{1}{3\sqrt{2}}, 0\right)$, we have $-\frac{1}{6\sqrt{2}x_1} = -\frac{1}{2}$, which gives us the

value of $x_1 = \frac{1}{3\sqrt{2}}$

Hence, $\beta = \frac{y_1}{2} = \frac{\sqrt{3}}{2}$

Solution 65:

Let point P be $(2t, t^2)$ and Q be (h, k)

$$h = \frac{2t}{3}, k = \frac{-2+t^2}{3}$$

$$\text{Hence locus is } 3k + 2 = \left(\frac{3h}{2}\right)^2 \Rightarrow 9x^2 = 12y + 8$$

Solution 66:

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$	$q \rightarrow (p \wedge (p \rightarrow q))$	$p \wedge q$
T	T	T	T	T	T	T
T	F	F	F	T	T	F
F	T	T	F	T	F	F
F	F	T	F	T	T	F

$p \vee (p \wedge q)$	$p \wedge q$	$p \wedge (p \wedge q)$
T	T	T
T	T	T
F	T	F
F	F	F

Solution 67:

$$\int_0^b \left(\sqrt{ax} - \frac{x^2}{a} \right) dx = \frac{a^2}{6} \Rightarrow \frac{2}{3} \sqrt{ab^3} - \frac{b^3}{3a} = \frac{a^2}{6} \quad \dots(1)$$

Since the area of $\Delta OQR = \frac{1}{2}$, we have the following.

$$\frac{1}{2} b^2 = \frac{1}{2} \Rightarrow b = 1$$

Substituting the value of b in equation (1), we get

$$4a\sqrt{a} - 2 = a^3 \Rightarrow a^6 + 4a^3 + 4 = 16a^3 \Rightarrow a^6 - 12a^3 + 4 = 0$$

Solution 66:

Let $z = \alpha$ and $z = \beta$ be the roots of the given equation.

Thus, we have $2\alpha = -b$ and $\alpha^2 + \beta^2 = 45$

$$(\alpha + 1)^2 + \beta^2 = 40, \text{ and hence, } (\alpha + 1)^2 - \alpha^2 = -5$$

$$\Rightarrow 2\alpha + 1 = -5 \Rightarrow 2\alpha = -6$$

Thus, $b = 6$ and $b^2 - b = 30$

Solution 69:

Consider $f(x) = \left(\frac{2^{1-x} + 2^{1+x} + 3^x + 3^{-x}}{2} \right)$

Using the relation $AM \geq GM$, we see that $f(x) \geq 3$

Solution 70:

Consider $D = \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 4 & 4 & 4 \end{vmatrix} = \begin{vmatrix} 3 & 4 & 5 \\ 1 & 2 & 3 \\ 0 & 0 & 0 \end{vmatrix} = 0$

By using $(R_3 \rightarrow R_3 - 2R_1 + 3R_2)$

Now let $P_3 \equiv 4x + 4y + 4z - \delta = 0$

If the system has any solutions, it will have infinitely many solutions.

So, $P_3 \equiv \alpha P_1 + \beta P_2$

This gives us $3\alpha + \beta = 4$ and $4\alpha + 2\beta = 4$

Solving these equations, we get $\alpha = 2$ and $\beta = -2$

So, for infinite solutions $2\mu - 2 = \delta$

Clearly, for $2\mu \neq \delta + 2$, the system inconsistent.

Solution 71:

Let $A = [a_{ij}]_{3 \times 3}$

Now, $\text{tr}(AA^T) = 3$, that is $a_{11}^2 + a_{12}^2 + a_{13}^2 + a_{21}^2 + \dots + a_{33}^2 = 3$

Possible cases are as follows.

$$\left. \begin{array}{ll} 0, 0, 0, 0, 0, 0, 1, 1 & \rightarrow 1 \\ 0, 0, 0, 0, 0, 0, -1, -1, -1 & \rightarrow 1 \\ 0, 0, 0, 0, 0, 0, 1, 1, -1 & \rightarrow 3 \\ 0, 0, 0, 0, 0, -1, 1, -1 & \rightarrow 3 \end{array} \right\}$$

So, number of ways is ${}^9C_6 \times 8 = 84 \times 8 = 672$

Solution 72:

Taking non-negative discriminant, we get the following

$$(a-10)^2 - 4(2)\left(\frac{33}{2} - 2a\right) \geq 0 \Rightarrow (a-10)^2 - 4(33-4a) \geq 0$$

$$\Rightarrow a^2 - 4a - 32 = 0 \Rightarrow a \in (-\infty, -4] \cup [8, \infty)$$

Solution 73:

The favourable cases are 0 red, 1 red, 2 red and 3 red.

Thus, the number of ways $= {}^7C_4 + {}^5C_1 \cdot {}^7C_3 + {}^5C_2 \cdot {}^7C_2 + {}^5C_3 \cdot {}^7C_1 = 35 + 175 + 210 + 70 = 490$

Solution 74:

$$\begin{aligned}
 \text{The given series} &= \sum_{k=1}^{20} \frac{k(k+1)}{2} \\
 &= \frac{1}{2} \sum_{k=1}^{20} (k^2 + k) \\
 &= \frac{1}{2} \left[\frac{20(21)(41)}{6} + \frac{20(21)}{2} \right] \\
 &= \frac{1}{2} \left[\frac{420 \times 41}{6} + \frac{20 \times 21}{2} \right] \\
 &= \frac{1}{2} [2870 + 210] \\
 &= 1540
 \end{aligned}$$

Solution 75:

Let $P \equiv (x_1, y_1)$

Thus, we have $2yy' - 6x = y' = 0 \Rightarrow y' = \left(\frac{6x_1}{1+2y_1} \right)$

That is, $\left(\frac{\frac{3}{2} - y_1}{-x_1} \right) = - \left(\frac{1+2y_1}{6x_1} \right)$, which gives us $9 - 6y_1 = 1 + 2y_1 \Rightarrow y_1 = 1$

$$\therefore x_1 = \pm 2$$

Hence, slope of tangent at the points $= \left(\pm \frac{12}{3} \right) = \pm 4$

So, we have $|n| = 4$